

13.10.2023

$$1) x_n \rightarrow l : \forall \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}:$$

$$|x_n - l| < \varepsilon, \forall n \geq N_\varepsilon$$

$$\frac{1}{\sqrt{n}} \rightarrow 0$$

Let $\varepsilon > 0$. We want to find $N_\varepsilon \in \mathbb{N}$ s.t.

$$\left| \frac{1}{\sqrt{n}} - 0 \right| < \varepsilon, \forall n \geq N_\varepsilon$$

$$\frac{1}{\sqrt{n}} < \varepsilon \Leftrightarrow \frac{1}{\varepsilon} < \sqrt{n} \Leftrightarrow \frac{1}{\varepsilon^2} < n$$

$$\text{Take } N_\varepsilon = \left\lceil \frac{1}{\varepsilon^2} \right\rceil + 1 > \frac{1}{\varepsilon^2}$$

$$\text{Then } \forall n \geq N_\varepsilon > \frac{1}{\varepsilon^2} \Leftrightarrow \frac{1}{\sqrt{n}} < \varepsilon; \forall n \geq N_\varepsilon$$

$$2) a) x_n = \sqrt{n+1} - \sqrt{n} = \frac{\cancel{n+1} - \cancel{n}}{\sqrt{n+1} + \sqrt{n}} =$$

$$= \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\infty} = 0$$

$$x_{n+1} \stackrel{?}{\leq} x_n$$

$$\frac{1}{\sqrt{n+2} + \sqrt{n+1}} \stackrel{?}{\leq} \frac{1}{\sqrt{n+1} + \sqrt{n}} \Leftrightarrow$$

$$\Leftrightarrow \sqrt{n+2} + \sqrt{n+1} \stackrel{?}{\geq} \sqrt{n+1} + \sqrt{n}$$

$$\Leftrightarrow \sqrt{n+2} \stackrel{?}{\geq} \sqrt{n} \quad \text{"True" (Decreasing)}$$

$$x_n \in (0, 1]$$

Bounded

$$b) x_n = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} =$$

$$= \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n} - \frac{1}{n+1}$$

$$= 1 - \frac{1}{n+1}$$

$$x_{n+1} - x_n \stackrel{?}{>} 0$$

$$1 - \frac{1}{n+2} - 1 + \frac{1}{n+1} \stackrel{?}{\geq} 0$$

$$\frac{1}{n+1} - \frac{1}{n+2} \stackrel{?}{>} 0 \text{ "True"}$$

$$x_{n+1} > x_n, \forall n \in \mathbb{N}$$

$$x_n < 1$$

$$x_n \in \left[\frac{1}{2}, 1 \right)$$

$$d) \quad x_n = \frac{2^n}{n!}$$

$$\frac{x_{n+1}}{x_n} = \frac{\cancel{2^{n+1}}^2}{\cancel{(n+1)!}^{n+1}} \cdot \frac{\cancel{n!}^1}{\cancel{2^n}_1} = \frac{2}{n+1} < 1 \quad \text{for } n+1 > 2 \Leftrightarrow n > 1$$

$$1, 2, 2, \frac{8}{6}$$

$$x_{n+1} < x_n, \text{ for } n > 1$$

$$\left. \begin{array}{l} x_n \in (0, 2] \rightarrow \text{bounded} \\ x_n \text{ decreasing, } n > 1 \end{array} \right\} \Rightarrow \text{Convergent}$$

Ratio test

$$\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = l \begin{cases} \rightarrow \text{if } l < 1, \text{ then } x_n \rightarrow 0 \\ \rightarrow \text{if } l > 1, \text{ then } x_n \rightarrow \infty \end{cases}$$

⊛ Behaves like a geometric prog.

In our case: $\frac{x_{n+1}}{x_n} = \frac{2}{n+1} \rightarrow 0 < 1 \Rightarrow$

$\Rightarrow x_n \rightarrow 0$

3) a) $x_n \sqrt{n} (\sqrt{n+1} - \sqrt{n})$

$$\begin{aligned} \lim_{n \rightarrow \infty} x_n &= \lim_{n \rightarrow \infty} \sqrt{n} (\sqrt{n+1} - \sqrt{n}) = \\ &= \lim_{n \rightarrow \infty} \frac{\sqrt{n} (n+1 - n)}{\sqrt{n+1} + \sqrt{n}} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \\ &= \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{n} (\sqrt{\frac{n+1}{n}} + 1)} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} = \\ &= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+1}} = \frac{1}{2} \end{aligned}$$

ⓧ $\lim_{n \rightarrow \infty} (2^n + 3^n)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} (3^n)^{\frac{1}{n}} \left[\left(\frac{2}{3} \right)^n + 1 \right]^{\frac{1}{n}} =$

$\downarrow \rightarrow 0$

$= 3 \cdot 1^0 = 3$

b) $(a_1^n + a_2^n + \dots + a_k^n)^{\frac{1}{n}}$, with $a_i > 0$

let $a_m = \max\{a_1, a_2, \dots, a_k\}$

$$(a_m)^{\frac{1}{n}} \left[\left(\frac{a_1}{a_m}\right)^n + \left(\frac{a_2}{a_m}\right)^n + \dots + \left(\frac{a_k}{a_m}\right)^n \right]^{\frac{1}{n}} \quad (**)$$

$$1 \leq E \leq k \quad \left(\frac{1}{n}\right) \underline{E}$$

$$1 \leq E^{\frac{1}{n}} \leq k^{\frac{1}{n}}$$

$$1 \leq \lim E \leq 1$$

$$\lim E = 1$$

(**)

$$a_m$$

c) $\sqrt[n]{n} = n^{\frac{1}{n}} = \infty^0$

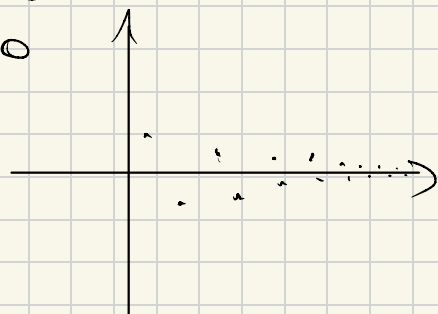
Hint $a = e^{\ln a}$, $f^g = e^{\ln f^g} = e^{g \cdot \ln f}$

$$n^{\frac{1}{n}} = e^{\frac{1}{n} \cdot \ln n} = e^{\frac{\ln n}{n}} = e^0 = 1, \quad n \rightarrow \infty$$

$$e) \lim_{n \rightarrow \infty} \frac{2^n + (-1)^n}{3^n} = \lim_{n \rightarrow \infty} \left[\frac{2^n}{3^n} + \frac{(-1)^n}{3^n} \right] =$$

$$= \lim_{n \rightarrow \infty} \left[\left(\frac{2}{3} \right)^n + \left(-\frac{1}{3} \right)^n \right] = 0$$

$$\left| -\frac{1}{3} \right|^n = \left(\frac{1}{3} \right)^n \rightarrow 0$$



$$f) \lim_{n \rightarrow \infty} \frac{(an+1)^2}{4n^2-2n+1}, \quad a \in \mathbb{R}$$

$$\lim_{n \rightarrow \infty} \frac{(an+1)^2}{4n^2-2n+1} = \lim_{n \rightarrow \infty} \frac{a^2 n^2 + 2an + 1}{4n^2 - 2n + 1} =$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 \left(a^2 + \frac{2a}{n} + \frac{1}{n^2} \right)}{n^2 \left(4 + \frac{-2}{n} + \frac{1}{n^2} \right)} = \frac{a^2}{4}, \quad \text{for } a \neq 0$$

$$a=0 \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{4n^2-2n+1} = \frac{1}{\infty} = 0, \quad \text{for } a=0$$

$$4) e_m = \left(1 + \frac{1}{n}\right)^n$$

$$(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots$$

$$+ \binom{n}{k}x^k + \dots + x^n$$

$$\binom{n}{k} = C_n^k = \frac{n!}{(n-k)!k!} = \frac{n(n-1)\dots(n-k+1)}{k!}$$

"n choose k"

$$e_m = 1 + \binom{n}{1}\frac{1}{n} + \binom{n}{2}\frac{1}{n^2} + \dots + \binom{n}{k}\frac{1}{n^k} + \dots + \frac{1}{n^n}$$

$$= 1 + 1 + \frac{n(n-1)}{2} + \dots + \frac{n(n-1)\dots(n-k+1)}{k!} \cdot \frac{1}{n^k} + \dots$$

$$+ \frac{1}{n^n}$$

$$= 1 + 1 + \frac{1}{2} \cdot \underbrace{1 \cdot \left(1 - \frac{1}{n}\right)}_{\geq 1} + \dots + \frac{1}{k!} \cdot \underbrace{1 \cdot \left(1 - \frac{1}{n}\right)}_{\geq 1} \cdot \dots$$

$$\cdot \underbrace{\left(1 - \frac{k-1}{n}\right)}_{\geq 1} + \dots + \frac{1}{n^n}$$

$$e_m < 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} + \dots + \frac{1}{n!}$$

$$\frac{1}{3!} = \frac{1}{2 \cdot 3} < \frac{1}{2^2}$$

$$\frac{1}{k!} = \frac{1}{2 \cdot 3 \cdot \dots \cdot k} < \frac{1}{2^{k-1}}$$

$$e_n < 1 + \underbrace{1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{k-1}} + \dots + \frac{1}{2^{n-1}}}_{\text{geometric series}}$$

$$e_n < 1 + \frac{1 - \frac{1}{2^n}}{1 - \frac{1}{2}} = 1 + 2\left(1 - \frac{1}{2^n}\right) < 1 + 2 = 3$$

$$\begin{aligned} &1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \\ &= \frac{1 - \frac{1}{2^n}}{1 - \frac{1}{2}} \end{aligned}$$

$$e_{n+k} = 1 + 1 + \frac{1}{2}\left(1 - \frac{1}{n+k}\right) + \dots + \frac{1}{k!} \cdot \left(1 - \frac{1}{k+1}\right)\left(1 - \frac{2}{k+1}\right) \dots \left(1 - \frac{k-1}{k+1}\right) + \dots$$

$$\Rightarrow \left. \begin{array}{l} e_{n+k} > e_n \\ e_n < 3 \end{array} \right\} \Rightarrow \exists \lim_{n \rightarrow \infty} e_n$$

$$5) a) \lim_{n \rightarrow \infty} \left(\frac{2n+1}{2n-1} \right)^n =$$

$$= \lim_{n \rightarrow \infty} \left(\frac{2n-1+2}{2n-1} \right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{2}{2n-1} \right)^n =$$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{2}{2n-1} \right)^{\frac{2n-1}{2} \cdot n \cdot \frac{2}{2n-1}} = \lim_{n \rightarrow \infty} \frac{2n}{2n-1} = e^1 = e$$

$$b) \lim_{n \rightarrow \infty} n(\ln(n+2) - \ln(n+1)) =$$

$$= \lim_{n \rightarrow \infty} n \left(\ln \frac{n+2}{n+1} \right) =$$

$$= \lim_{n \rightarrow \infty} \ln \left(\frac{n+2}{n+1} \right)^n = \ln \lim_{n \rightarrow \infty} \left(\frac{n+1+1}{n+1} \right)^n$$

$$= \ln \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n+1} \right)^n = \ln \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n+1} \right)^{(n+1) \cdot \frac{1}{n+1}} =$$

$$= \ln e^{\lim_{n \rightarrow \infty} \frac{1}{n+1}} = \ln e^1 = \ln e = 1$$

$$7) a) \lim_{n \rightarrow \infty} \frac{x_1 + \dots + x_n}{n} \stackrel{S-C}{=} \lim_{n \rightarrow \infty} \frac{x_{n+1}}{1} =$$

$$= \lim_{n \rightarrow \infty} x_n$$

(ex) $x_n = (-1)^n$, $x_1 + x_2 + \dots + x_n \in \{0, -1\}$

x_n overall converges but the sequence does not!

$$b) \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \dots + \frac{1}{n}}{n} \xrightarrow{S-C} \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \dots + \frac{1}{n}}{\ln n} \xrightarrow{S-C} \lim_{n \rightarrow \infty} \frac{\frac{1}{n+1}}{\ln \frac{n+1}{n}} =$$

$$= \lim_{n \rightarrow \infty} \frac{1}{(n+1) \ln \frac{n+1}{n}} = 1$$